Fast-forward a quarter century, to May 1985: a phone call from Carl Sagan asking me to critique the relativistic science in his forthcoming novel *Contact*. I happily agreed. We were close friends, I thought it would be fun, and, besides, I still owed him one for introducing me to Lynda Obst.

Carl sent me his manuscript. I read it and I loved it. But there was one problem. He sent his heroine, Dr. Eleanor Arroway, through a black hole from our solar system to the star Vega. But I knew that a black-hole interior *cannot* be a route from here to Vega or to anywhere else in our universe. After plunging through the black hole's horizon, Dr. Arroway would get killed by its singularity. To reach Vega fast, she needed a wormhole, not a black hole. But a wormhole that does *not* pinch off. A *traversable* wormhole.

So I asked myself, What do I have to do to Flamm's wormhole to save it from pinching off; to hold it open, so it can be traversed? A simple thought experiment gave me the answer.

Suppose you have a wormhole that is spherical like Flamm's, but unlike Flamm's it does not pinch off. Send a light beam into the wormhole, radially. Since all the beam's light rays travel radially, the beam must have the shape shown in Figure 14.4. It is converging (its cross-sectional area is decreasing) as it enters the wormhole, and it is diverging (its area is increasing) as it leaves the wormhole. The wormhole has bent the light rays outward, as would a diverging lens.